

Recommendations — ulms-highgrade-pulm-mets-tp53-smarc b1-hla-a2-w9t4

Ranked therapeutic options with likelihood, toxicity, mechanism risk, and key references

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Recommendations — ulms-highgrade-pulm-mets-tp53-smarcb1-h

Profile snapshot

- **Primary site:** uterus (myometrium)
- **Histology:** high-grade uterine leiomyosarcoma (uLMS), spindle-cell morphology; diagnosis confirmed on both the pulmonary biopsy and the re-reviewed uterine specimen by a reference pathology laboratory (August 2026)
- **Stage:** metastatic (stage IV): bilateral pulmonary metastases. Radiologic read by the consulting sarcoma oncologist: lesions unusually infiltrative and possibly inflammatory in appearance. Primary resected September 2024 (signed out benign at the time); metastatic diagnosis July-August 2026
- **ECOG:** 1
- **Age band:** 30-39
- **Sex:** female
- **Biomarkers:** ER (estrogen receptor) <1% positive (decision-relevant negative); PR (progesterone receptor) <1% positive (decision-relevant negative); SMARCB1 G344Afs13 (*frameshift, called pathogenic*) detected on plasma ctDNA (Guardant liquid biopsy, August 2026). NOT yet confirmed on tumor tissue. A truncating SMARCB1 alteration raises the hypothesis of SMARCB1/INI1 deficiency, but SMARCB1 loss is uncommon in uterine leiomyosarcoma and a plasma-only call is a hypothesis, not an established tumor feature. (ihc_pending); TP53 Q52 (nonsense, called pathogenic) detected on plasma ctDNA (Guardant liquid biopsy, August 2026). NOT tissue-confirmed. Carries an explicit clonal-haematopoiesis (CHIP) caveat: TP53 is the most common CHIP gene, and a ctDNA-only TP53 call in a patient with potentially low tumor fraction can originate from blood rather than tumor. TP53 alteration is also expected in the large majority of uLMS, so if tumor-derived it is unsurprising and largely prognostic rather than directly actionable. (ngs_pending); HLA class I typing (allele-level) HLA-A02:01, A03:01; HLA-B39:01, B44:02; HLA-C05:01, C12:03. HLA-A02:01 is present: this is the restriction allele required by the MAGE-A4 and NY-ESO-1 TCR-engineered T-cell programs. Typing alone is NOT eligibility; those programs also require tumor expression of the antigen by IHC, which has not been done.; MAGE-A4 expression not tested. Newly decision-relevant because HLA-A02:01 is confirmed present: MAGE-A4 expression by IHC is the second, still-missing half of eligibility screening for A02:01-restricted MAGE-A4 TCR-T programs (afamitresgene autoleucel is approved in synovial sarcoma; uLMS access would be via trial). (ihc_pending); NY-ESO-1 expression not tested. Newly decision-relevant because HLA-A02:01 is confirmed present: NY-ESO-1 expression by IHC is the second, still-missing half of eligibility screening for A02:01-restricted NY-ESO-1 TCR-T programs (trial-only). (ihc_pending); PRAME expression not tested. Same type-first-stain-second dependency: HLA-A02:01 is present, and PRAME-directed TCR/ImmTAC programs (trial-only) require demonstrated PRAME expression. (ihc_pending); Guardant ctDNA variants of uncertain significance (recorded, not pursued) Four VUS at 0.09-1.1% VAF, at or near the assay limit of detection: ARAF L161Ffs16 (0.09%), ERBB3 G914E (0.9%), GREM1 L14Sfs70 (0.1%), P2RY8 R234S (1.1%). Two are frameshifts in genes with no established role in this histology. Treated as low-confidence subclonal or technical findings, not targetable features. Explicitly: ERBB3 G914E at 0.9% VAF is NOT a HER-family targeting rationale. (unknown); RB1 no alteration reported on the plasma ctDNA assay; a plasma negative is not exclusionary (possible low tumor shedding, panel coverage limits). Tissue-level status remains unmeasured. (ngs_pending); ATRX not reported on the plasma ctDNA assay (coverage uncertain); tissue-level status remains unmeasured (ngs_pending); MED12 not reported on the plasma ctDNA assay (coverage uncertain); tissue-level status remains unmeasured (ngs_pending); Homologous recombination status incl. BRCA2 not settled by the Guardant plasma panel (no BRCA1/2 alteration reported, but plasma sensitivity is limited at low tumor fraction and HRD signatures require tissue). Remains unmeasured at decision-relevant resolution.

(ngs_pending); NTRK fusion not settled by the Guardant plasma panel (fusion detection in plasma is insensitive at low tumor fraction). Remains unmeasured at decision-relevant resolution. (ngs_pending); TMB (tumor mutational burden) not reported on the Guardant assay; remains unmeasured. Plasma TMB at low tumor fraction is unreliable in any case. (ngs_pending); MSI / MMR not settled by the Guardant plasma panel; remains unmeasured (ngs_pending); PD-L1 not tested (not assessable on a plasma assay) (ihc_pending)

- **Efficacy/toxicity weight:** 0.7

44 ranked therapeutic options across 11 targetable features. Biomarker workup steps live in the standalone Target validation paths report.

Hrd Brca interventions

1. Olaparib + temozolomide (NCI Protocol 10250 regimen in uterine leiomyosarcoma), post-induction, with tissue HRD/HRR testing ordered now

Access: Off-label

Likelihood of effect: Moderate in her exact histology: ORR 27% (6 of 22) and mPFS 6.9 months in advanced uterine LMS, prespecified endpoint met (pmid:37467452).

Toxicity burden: High (G3/4 neutropenia 75%, thrombocytopenia 32%, anaemia; managed with dose modification, no treatment-related deaths)

Counter-productive MoA:Low (Overlapping myelosuppression after LMS-04 may force dose reductions that cost the PARP exposure the regimen depends on.)

Overall:The best lineage-matched targeted signal in this file, sitting one line away and worth ordering tissue HRD for now — at a heavy second consecutive marrow cost.

Key references: PMID 37467452 · NCT03880019 · NCT05432791

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

Fluzoparib + camrelizumab with concurrent SBRT to lung metastases (NCT06074692) — _Not enrollable_

Access: Not yet accessible

Likelihood of effect: Not enrollable: Shanghai single site with two drugs unavailable in the US. Local SBRT carries its own retrospective evidence (pmid:37532104).

Toxicity burden: High (PARP-inhibitor myelosuppression plus checkpoint toxicity and concurrent radiotherapy; no combination profile published)

Counter-productive MoA: Moderate (Concurrent radiotherapy with a PARP inhibitor raises pneumonitis risk in already-infiltrated lung.)

Overall: The one protocol combining her best mechanism with her locoregional question, in Shanghai, with two drugs the US does not have.

Key references: NCT06074692 · PMID 37532104

PARP inhibition in BRCA2-altered uterine leiomyosarcoma (genomic series) — _Consolidated_

Access: Off-label

Likelihood of effect: Consolidated into rank 2. On its own: four of four with durable benefit in a tier-4 retrospective series (pmid:30541756), hypothesis-generating only.

Toxicity burden: Low (no systematic AE reporting in a four-patient off-label series; class profile carried on the rank-2 row)

Counter-productive MoA: Low (No mechanism objection; the risk is over-reading a four-patient series as a selection rule.)

Overall: The reason BRCA2 belongs on her tissue order, and a four-patient series that cannot carry more weight than that.

Key references: PMID 30541756 · PMID 37143137 · PMID 37467452

Evidence in detail

Olaparib + temozolomide (NCI Protocol 10250 regimen in uterine leiomyosarcoma), post-induction, with tissue HRD/HRR testing ordered now

[Ingham et al. J Clin Oncol 2023](#)

- **Disease context:** advanced uterine leiomyosarcoma after progression on ≥ 1 prior line (2L+) — 22 evaluable uLMS patients; 59% had ≥ 3 prior lines; HRD not required for entry but assessed on paired biopsies (RAD51 assay, WES/RNAseq)
- **n:** 22
- **Toxicities:** neutropenia (grade ≥ 3): 75% (n=22); thrombocytopenia (grade ≥ 3): 32% (n=22)
- **Efficacy:** ORR 23%; ORR 27%; PFS 6.9 months; PFS 11.2 months; biomarker 50%

Fluzoparib + camrelizumab with concurrent SBRT to lung metastases (NCT06074692)

[Lebow et al. Radiother Oncol 2023](#)

- **Disease context:** sarcoma pulmonary metastases, including patients unfit for or declining resection (any) — 66 consecutive patients, median follow-up 36 months; oligometastatic and intrathoracic-only disease predicted longer survival
- **n:** 66
- **Toxicities:** pneumonitis (grade 1-2): 5% (n=66); esophagitis (grade 1-2): 2% (n=66); any adverse event (grade ≥ 3): 0% (n=66)

- **Efficacy:other** 3.1% local failure (95% CI 0.9-10.6); **other** 7.4% local failure (95% CI 4.0-13.9); **OS_rate** 74% (95% CI 64-86); **OS_rate** 49% (95% CI 38-63)

PARP inhibition in BRCA2-altered uterine leiomyosarcoma (genomic series)

****Seligson et al. *Oncologist* 2019****

- **Disease context:**leiomyosarcoma; **BRCA2-altered uterine LMS subset** (any) — 2,548 STS genomic profiles across two cohorts; BRCA2 alterations concentrated in uLMS (10% of uLMS); 4 uLMS patients with functional BRCA2 loss treated with PARP inhibitors
- **n:** 4
- **Toxicities:** No systematic AE reporting; four-patient off-label series.
- **Efficacy:other** 4/4 durable clinical benefit; **biomarker** 10%

****Ingham et al. *J Clin Oncol* 2023****

- **Disease context:**advanced uterine leiomyosarcoma after progression on ≥ 1 prior line (2L+) — 22 evaluable uLMS patients; 59% had ≥ 3 prior lines; HRD not required for entry but assessed on paired biopsies (RAD51 assay, WES/RNAseq)
- **n:** 22
- **Toxicities:**neutropenia (grade ≥ 3): 75% (n=22); thrombocytopenia (grade ≥ 3): 32% (n=22)
- **Efficacy:**ORR 23%; **ORR** 27%; **PFS** 6.9 months; **PFS** 11.2 months; **biomarker** 50%

Rb1 Status interventions

1. Abemaciclib + gemcitabine and docetaxel (NCT06498648, MD Anderson randomised phase 1/2), Rb-intact required

Access: Trial only

Likelihood of effect: Unknown for the add-on: no abemaciclib efficacy data in LMS. The floor is the randomised gem/doc control arm, ORR 35.8% first-line uterine (pmid:18534250).

Toxicity burden: Moderate (backbone G3 anaemia 24%, G3 fatigue 17%, G3-4 neutropenia 17%, one G4 pulmonary event in 42; abemaciclib diarrhoea uncharacterised on this doublet)

Counter-productive MoA:Moderate (CDK4/6-induced G1 arrest can antagonise S-phase-dependent gemcitabine; Rb-intact status, the premise of the whole seat, is unmeasured.)

Overall:The one investigational seat that survives LMS-04 instead of competing with it, routed by a single Rb stain and carrying no efficacy data of its own.

Key references: NCT06498648 · PMID 18534250 · PMID 28882536

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no

peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

REC-617 (selective CDK7 inhibitor) in RB1-negative leiomyosarcoma (NCT07633756, MD Anderson) — Thin evidence

Access: Trial only

Likelihood of effect: Unknown: 15-patient phase 1b with a safety primary endpoint and no efficacy readout.

Toxicity burden: Moderate (no published sarcoma toxicity profile; CDK7-class myelosuppression and gastrointestinal toxicity expected)

Counter-productive MoA:Low (No mechanism objection raised; the risk is a null first-in-class readout rather than antagonism.)

Overall:The Rb-loss half of a single stain that also routes the abemaciclib seat — free to answer now, with no efficacy data behind it.

Key references: NCT07633756 · NCT06498648

Evidence in detail

Abemaciclib + gemcitabine and docetaxel (NCT06498648, MD Anderson randomised phase 1/2), Rb-intact required

Hensley et al. *Gynecol Oncol* 2008

- **Disease context:**metastatic uterine leiomyosarcoma, chemotherapy-naïve (1L) — 42 women with advanced uLMS, no prior chemotherapy; 39 evaluable for response
- **n:** 42
- **Toxicities:**neutropenia (grade 3): 5% (n=42); neutropenia (grade 4): 12% (n=42); anemia (grade 3): 24% (n=42); thrombocytopenia (grade 3): 9.5% (n=42); fatigue (grade 3): 17% (n=42); pulmonary toxicity (grade 4): 2.4% (n=42)
- **Efficacy:**ORR 35.8% (95% CI 23.5-49.6); DCR 62%; PFS 4.4 months; OS 16 months; DoR 6 months

Seddon et al. *Lancet Oncol* 2017

- **Disease context:**previously untreated advanced or metastatic soft-tissue sarcoma (1L) — Trojani grade 2-3 STS; leiomyosarcoma (uterine and non-uterine) was the largest histology stratum
- **n:** 257
- **Toxicities:**neutropenia (grade ≥3): 20% (n=126); febrile neutropenia (grade ≥3): 12% (n=126); fatigue (grade ≥3): 14% (n=126); oral mucositis (grade ≥3): 2% (n=126)
- **Efficacy:**PFS_rate 46.4% (95% CI 37.5-54.8); PFS_rate 46.3% (95% CI 37.5-54.6); HR_PFS 1.28 (95% CI 0.99-1.65)

Prame lhc interventions

1. IMA203 / IMA203CD8 PRAME-directed TCR-T cells (NCT03686124), sponsor screening

started in the first-line window

Access: Trial only

Likelihood of effect: Low-to-uncertain: 28.9% confirmed ORR in a 41-patient phase 1 with no uLMS patient and mDoR 4.4 months (pmid:40205198).

Toxicity burden: Moderate (severe CRS 4.9%, no severe ICANS in n=41; lymphodepletion cytopenias expected but not tabulated per term)

Counter-productive MoA: **Moderate** (Antigen-loss or HLA-A*02:01 LOH escape; a local PRAME stain has no established operating characteristics against the sponsor assay.)

Overall: The only cellular axis her confirmed HLA-A*02:01 opens, double-gated on a sponsor PRAME assay and a standard-of-care-exhaustion clause.

Key references: PMID 40205198 · NCT03686124 · PMID 40152485

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

Brenetafusp (IMC-F106C, PRAME x CD3 ImmTAC) (NCT04262466) — _Not enrollable_

Access: Trial only

Likelihood of effect: Not enrollable: accrual closed, no peer-reviewed clinical publication for the agent; class precedent is uveal-melanoma-specific (pmid:34551229).

Toxicity burden: Moderate (ImmTAC-class rash, pyrexia and pruritus front-loaded over the first doses; class discontinuation for AE 2%)

Counter-productive MoA: **Low** (Antigen-loss and HLA LOH escape possible; no mechanism objection raised.)

Overall: The off-the-shelf PRAME route that needs no cell manufacturing, closed to accrual — one phone call away from being worth re-checking.

Key references: NCT04262466 · PMID 34551229

Evidence in detail

IMA203 / IMA203CD8 PRAME-directed TCR-T cells (NCT03686124), sponsor screening started in the first-line window

Wermke et al. *Nat Med* 2025

- **Disease context:** PRAME-positive recurrent/refractory solid tumours in HLA-A*02-positive patients (melanoma-dominant; sarcoma included) (2L+) — 41 patients started lymphodepletion, 40 treated; PRAME

expression and HLA-A*02 required; confirmed responses enriched at higher dose and higher PRAME expression

- n: 41
- **Toxicities:**cytokine release syndrome (grade ≥ 3): 4.9% (n=41); **neurotoxicity (ICANS)** (grade ≥ 3): 0% (n=41)
- **Efficacy:**ORR 28.9%; **ORR** 52.5%; **DoR** 4.4 months

Brenetafusp (IMC-F106C, PRAME x CD3 ImmTAC) (NCT04262466)

****Nathan et al. *N Engl J Med* 2021****

- **Disease context:**previously untreated metastatic uveal melanoma, **HLA-A*02:01-positive** (1L) — 378 HLA-A*02:01 patients randomised 2:1 vs investigator's choice (pembrolizumab, ipilimumab, or dacarbazine)
- n: 378
- **Toxicities:**rash (grade any): 83% (n=252); **pyrexia** (grade any): 76% (n=252); **pruritus** (grade any): 69% (n=252); **treatment discontinuation for AE** (grade any): 2% (n=252)
- **Efficacy:**HR_OS 0.51 (95% CI 0.37-0.71); **HR_PFS** 0.73 (95% CI 0.58-0.94)

Pulmonary Oligometastatic interventions

1. In vivo lung perfusion with doxorubicin plus metastasectomy for sarcoma pulmonary metastases (NCT02811523, Toronto), as post-induction consolidation

Access: Trial only

Likelihood of effect: Uncertain: metastasectomy series give 55.4% five-year OS in resected uterine sarcoma (pmid:37063124), but perfusion itself is phase 1 with no human efficacy readout.

Toxicity burden: Moderate (thoracotomy morbidity; acute lung injury is the trial's registered primary endpoint; no published human safety series in this population)

Counter-productive MoA:Moderate (Perfusion injury to already-infiltrated lung; and a full induction spends the anthracycline budget that keeps the door open.)

Overall:The one option in this dossier aimed at cure rather than control, and the one whose anatomic premise her infiltrative bilateral pattern may defeat.

Key references: NCT02811523 · PMID 37063124 · PMID 26952295

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

Pulmonary suffusion with cisplatin plus metastasectomy for sarcoma lung metastases (NCT03965234, Roswell Park) — _Thin evidence_

Access: Trial only

Likelihood of effect: Unknown: phase 1/2 with a local-toxicity endpoint; the survival rationale belongs to metastasectomy series, not to suffusion (pmid:9011700).

Toxicity burden: Moderate (thoracotomy morbidity plus regional cisplatin exposure; no published outcome series)

Counter-productive MoA: **Moderate** (Same resectability premise as the Toronto protocol; an infiltrative bilateral pattern may never qualify.)

Overall: The US locoregional trial with no prior-line requirement, resting on a resectability judgement her current imaging may not support.

Key references: NCT03965234 · PMID 9011700

Evidence in detail

In vivo lung perfusion with doxorubicin plus metastasectomy for sarcoma pulmonary metastases (NCT02811523, Toronto), as post-induction consolidation

Nobori et al. *JTCVS Open* 2023

- **Disease context:** lung metastases from uterine primaries; sarcoma subset n=46 of 319 (any) — Metastatic Lung Tumor Study Group of Japan, 1984-2016; risk-adjusted analysis by primary histology
- **n:** 319
- **Toxicities:** Operative complications not reported in the abstract.
- **Efficacy:** OS_rate 55.4%; other DFI >=12 months prognostic

Reck Dos Santos et al. *Ann Thorac Surg* 2016

- **Disease context:** —
- **n:** —
- **Toxicities:** —
- **Efficacy:** —

Pulmonary suffusion with cisplatin plus metastasectomy for sarcoma lung metastases (NCT03965234, Roswell Park)

Pastorino et al. *J Thorac Cardiovasc Surg* 1997

- **Disease context:** lung metastases from any primary; sarcoma 2173 of 5206 cases (any) — 5206 metastasectomies from 18 thoracic units; prognostic model built on resectability, disease-free interval, and number of lesions
- **n:** 5206
- **Toxicities:** Operative morbidity and mortality not reported in the abstract; the registry's message is selection: completeness of resection, DFI and lesion count drive outcome.
- **Efficacy:** OS_rate 36%; OS_rate 13%; OS_rate 45%; OS_rate 27%

Ulms Histology interventions

1. Ivonescimab (PD-1 x VEGF bispecific), MSK single-arm phase 2 in leiomyosarcoma (NCT07516925)

Access: Trial only

Likelihood of effect: Low: no sarcoma efficacy data anywhere, and both component axes already failed in uLMS (pmid:25713428, pmid:28440953). HARMONi-2 is an NSCLC result.

Toxicity burden: Moderate (G3+ treatment-related AEs 29% in NSCLC, VEGF-class hypertension and bleeding, G3+ immune-related AEs 7%)

Counter-productive MoA: **Moderate** (Both mechanisms have already been beaten in this disease; a spent first-line window is the counter-productive outcome.)

Overall: The only seat reachable before cycle 1 by explicit protocol language, and the one four of five personas refused on three mutually independent negative trials.

Key references: NCT07516925 · PMID 40057343 · PMID 25713428

2. Nivolumab monotherapy in uterine leiomyosarcoma

Access: Off-label

Likelihood of effect: Low for response: ORR 0% with mPFS 1.8 months in the only dedicated uLMS PD-1 trial, stopped for futility (pmid:28440953).

Toxicity burden: Moderate (G3+ events in 9 of 12, three attributed to drug: abdominal pain, amylase/lipase rise, fatigue)

Counter-productive MoA: **Moderate** (Prior PD-1 exposure permanently closes the ivonescimab seat and buys nothing this histology has shown it responds to.)

Overall: The negative result that defines the checkpoint question in this disease: 0 of 12, stopped for futility.

Key references: PMID 28440953 · PMID 28988646

3. Nivolumab +/- ipilimumab, dual checkpoint blockade in sarcoma (Alliance A091401)

Access: Off-label

Likelihood of effect: Low-to-uncertain: ORR 16% across sarcoma histologies with no uLMS breakdown, against a 0-of-12 and 0-of-10 record in this disease (pmid:29370992).

Toxicity burden: Moderate (G3+ anaemia 19%, hypotension 10%; treatment-related serious AEs 26%; no treatment-related deaths)

Counter-productive MoA: **Moderate** (Repeats a mechanism this histology has twice defeated, and forecloses the ivonescimab seat on the way.)

Overall:The evidence that dual blockade beats single-agent PD-1 in sarcoma generally, with nothing in it specific to uterine leiomyosarcoma.

Key references: PMID 29370992 · PMID 28440953

4. Doxorubicin + pembrolizumab, first-line chemo-immunotherapy (phase 1/2)

Access: Off-label

Likelihood of effect: Low for response: ORR 19% with the primary endpoint missed, durable responses clustered outside LMS (pmid:32910151).

Toxicity burden: Moderate (no dose-limiting toxicities reported; anthracycline myelosuppression plus immune-related events, no per-term G3 table published)

Counter-productive MoA:**Moderate** (Spends the first-line window and the anthracycline budget on a mechanism this histology has not responded to.)

Overall:Deliverable first-line chemo-immunotherapy whose response endpoint failed and whose responders were other histologies.

Key references: PMID 32910151 · PMID 35835135

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

Peposertib (DNA-PK inhibitor) + low-dose pegylated liposomal doxorubicin (NCT05711615), anthracycline-ceiling-gated — _Thin evidence_

Access: Trial only

Likelihood of effect: Unknown: recruiting phase 1 with a dose-finding endpoint and no efficacy readout in any histology.

Toxicity burden: Moderate (DNA-PK inhibitor plus anthracycline; myelosuppression and mucositis expected, no published grade profile)

Counter-productive MoA:**Low** (Radiosensitiser-class mechanism with no antagonism vector raised; the risk is a spent anthracycline budget.)

Overall:A later-line seat whose 360 mg/m² anthracycline ceiling makes it an induction-planning question rather than a progression question.

Key references: NCT05711615 · PMID 35835135

ADCE-D01 (uPARAP/Endo180-directed antibody-drug conjugate) in soft-tissue sarcoma (NCT06797999) — _Thin evidence_

Access: Trial only

Likelihood of effect: Unknown: first-in-human dose-escalation with no efficacy readout.

Toxicity burden: Moderate (ADC-class myelosuppression and payload effects; no published grade profile)

Counter-productive MoA:Low (No mechanism objection raised; risk is a null first-in-human readout.)

Overall:A sarcoma-dedicated ADC with a line window that LMS-04 opens and a third line would close — worth a calendar entry, not a rank.

Key references: NCT06797999

ZL-6201 (LRRC15-directed antibody-drug conjugate) in sarcoma (NCT07374848) — _Thin evidence_

Access: Trial only

Likelihood of effect: Unknown: phase 1 dose-escalation with no efficacy readout.

Toxicity burden: Moderate (ADC-class myelosuppression and payload effects; no published grade profile)

Counter-productive MoA:Low (No mechanism objection raised; risk is a null first-in-human readout.)

Overall:A post-induction ADC seat with the strictest performance-status ceiling in this tier and no efficacy data.

Key references: NCT07374848

Gemcitabine + nab-sirolimus in leiomyosarcoma or TSC1/TSC2-altered sarcoma (NCT06308419) — _Thin evidence_

Access: Trial only

Likelihood of effect: Unknown: 18-patient phase 1 with a safety endpoint and no efficacy readout.

Toxicity burden: Moderate (gemcitabine myelosuppression plus mTOR-inhibitor stomatitis and metabolic effects; no published combination profile)

Counter-productive MoA:Low (Gemcitabine exposure here permanently forecloses the abemaciclib randomised seat — a sequencing cost, not an antagonism.)

Overall:A post-induction phase 1 whose gemcitabine backbone would permanently close the one trial seat LMS-04 preserves.

Key references: NCT06308419

All-trans retinoic acid + cemiplimab in leiomyosarcoma after standard chemotherapy (NCT06528769) — _Thin evidence_

Access: Trial only

Likelihood of effect: Low: 16-patient single-arm phase 2 with no readout, built on a checkpoint component that went 0 of 12 in dedicated uLMS (pmid:28440953).

Toxicity burden: Moderate (immune-related toxicity plus retinoid effects — hypertriglyceridaemia, mucocutaneous dryness, headache; no published combination profile)

Counter-productive MoA: **Moderate** (Single-agent PD-1 has failed twice in this histology; the retinoid rationale is preclinical and PRAME-dependent.)

Overall: The one PRAME-adjacent rationale that does not need a cell product, attached to a checkpoint backbone this disease has already beaten twice.

Key references: NCT06528769 · PMID 28440953

**FAP-directed radioligand therapy ([²²⁵Ac]RTX-2358, [²¹²Pb]Pb-PSV359, [¹⁷⁷Lu]Lu-FAP-2286) —
Thin evidence**

Access: Trial only

Likelihood of effect: Unknown: three phase 1 or 1/2 programmes with safety endpoints and no sarcoma efficacy readout.

Toxicity burden: Moderate (radioligand marrow suppression and renal exposure; no sarcoma-specific grade profile published)

Counter-productive MoA: **Low** (Stromal FAP targeting has no antagonism vector raised; the risk is absent uptake.)

Overall: The one axis whose screening test is a scan rather than a stain, with no sarcoma efficacy data behind any of its three programmes.

Key references: NCT07156565 · NCT06710756 · NCT04939610

**Zanzalintinib (XL092), next-generation multikinase inhibitor with a leiomyosarcoma cohort (NCT06571734) —
Not enrollable**

Access: Trial only

Likelihood of effect: Unknown: phase 2 with no readout. Class benchmark pazopanib gives roughly three months of PFS with no OS gain (pmid:22595799).

Toxicity burden: Moderate (VEGFR-TKI class: hypertension, fatigue, diarrhoea, hand-foot syndrome; no agent-specific sarcoma profile)

Counter-productive MoA: **Low** (Antiangiogenic wound-healing impairment would conflict with any planned metastasectomy; sequencing issue, not antagonism.)

Overall: A third-line seat at the earliest, worth naming now only because its pathology-review requirement takes months to arrange.

Key references: NCT06571734 · PMID 22595799

Lurbinectedin + doxorubicin, first-line anthracycline-naïve leiomyosarcoma (SaLuDo, NCT06088290) — _Not enrollable_

Access: Off-label

Likelihood of effect: Unknown: phase 3 accrued and unreported; no efficacy data public.

Toxicity burden: Moderate (trabectedin-class myelosuppression and transaminitis expected; no combination profile published)

Counter-productive MoA:Low (Mechanistically redundant with the trabectedin already planned in induction.)

Overall:The trial that matched her indication and line exactly and closed before her diagnosis was corrected — a read-out to watch, not a seat.

Key references: NCT06088290

Pegylated liposomal doxorubicin maintenance after first-line chemotherapy (MELODY, NCT06981637) — _Not enrollable_

Access: Off-label

Likelihood of effect: Unknown: phase 2 with no readout, and the same question is answered inside her planned regimen (pmid:35835135).

Toxicity burden: Moderate (liposomal anthracycline: palmar-plantar erythrodysesthesia, mucositis, myelosuppression)

Counter-productive MoA:Low (Further anthracycline exposure erodes the cumulative-dose headroom that two other protocols gate on.)

Overall:A maintenance question her planned regimen already answers, in a trial she cannot reach.

Key references: NCT06981637 · PMID 35835135

Anlotinib (catequentinib, AL3818) in refractory soft-tissue sarcoma (APROMISS, NCT03016819) — _Not enrollable_

Access: Not yet accessible

Likelihood of effect: Low-to-uncertain: ORR 13% overall with an 11-month mPFS in a 26-patient LMS subset that was not prespecified for inference (pmid:29895706).

Toxicity burden: Low (G3+ hypertension 4.8%, triglyceride elevation 3.6%, pneumothorax 2.4%; no treatment-related deaths)

Counter-productive MoA:Moderate (Pneumothorax risk from tumour cavitation in bilateral lung metastases; wound-healing impairment conflicts with metastasectomy.)

Overall:The best-looking VEGFR-TKI subset number in the file, from a non-prespecified 26-patient subgroup, in an arm that is currently suspended.

Key references: PMID 29895706 · NCT03016819 · PMID 39918552

MASCT-I (autologous dendritic cell and cytokine-induced killer cell therapy) + doxorubicin + ifosfamide, first line (NCT06277154) — _Not enrollable_

Access: Not yet accessible

Likelihood of effect: Not enrollable: mainland China single site with locally manufactured cells; no efficacy data for the cellular component.

Toxicity burden: High (backbone febrile neutropenia 46% in EORTC 62012 despite pegfilgrastim; cellular component profile unpublished)

Counter-productive MoA:Low (No mechanism objection raised; the barrier is geography and manufacturing.)

Overall:One of the few first-line investigational options anywhere, single-site in China with the cell product made locally.

Key references: NCT06277154 · PMID 24618336

Tislelizumab + liposomal doxorubicin and ifosfamide, first line (NCT06849986) — _Not enrollable_

Access: Not yet accessible

Likelihood of effect: Not enrollable: Shanghai single site. The nearest published analogue missed its response endpoint (ORR 19%, pmid:32910151).

Toxicity burden: High (anthracycline-ifosfamide backbone toxicity plus immune-related events; no combination profile published)

Counter-productive MoA:Moderate (Checkpoint blockade has failed twice in this histology; adding it to chemotherapy repeats a tested idea.)

Overall:A first-line chemo-immunotherapy seat in Shanghai, on a mechanism this disease has already defeated twice.

Key references: NCT06849986 · PMID 32910151

[177Lu]Lu-ITG-PSMA-1 with [68Ga]Ga-PSMA-11 PET selection (NCT05420727) — _Unavailable_

Access: Unavailable

Likelihood of effect: Not available: completed single-centre phase 1, no successor study registered.

Toxicity burden: Moderate (radioligand marrow and renal exposure; no published grade profile from the 14-patient study)

Counter-productive MoA:Low (Neovascular rather than tumour-cell targeting; no mechanism objection raised.)

Overall:A finished single-centre Swiss phase 1 with no successor — the dead half of the radioligand branch.

Key references: NCT05420727

Checkpoint-blockade target biology in uterine leiomyosarcoma (immune classes, tertiary lymphoid structures, pS6-high sensitisation) — _Consolidated_

Access: Off-label

Likelihood of effect: Low: mechanistic rationale exists but the clinical record in this histology is 0 of 10 and 0 of 12 (pmid:28988646, pmid:28440953).

Toxicity burden: Low (none — this row is target biology; toxicity is carried on the agent rows)

Counter-productive MoA: **Moderate** (Sensitisation hypotheses have no available assay, and the histology has twice defeated the unmodified mechanism.)

Overall: The mechanism row behind five checkpoint options, and the reason the immune-phenotype stains ride the rank-1 block cut.

Key references: PMID 28988646 · PMID 28440953

Evidence in detail

Ivonescimab (PD-1 x VEGF bispecific), MSK single-arm phase 2 in leiomyosarcoma (NCT07516925)

****Xiong et al. *Lancet* 2025****

- **Disease context:** **PD-L1-positive advanced NSCLC, first line (1L)** — 398 patients in China; PD-L1 TPS $\geq 1\%$; EGFR/ALK excluded
- **n:** 398
- **Toxicities:** **any treatment-related AE** (grade ≥ 3): 29% (n=198); **immune-related AE** (grade ≥ 3): 7% (n=197)
- **Efficacy:** **HR_PFS** 0.51 (95% CI 0.38-0.69); **PFS** 11.1 months; **PFS** 5.8 months; **TRAE_rate** 29%

****Hensley et al. *J Clin Oncol* 2015****

- **Disease context:** **chemotherapy-naive metastatic unresectable uterine leiomyosarcoma (1L)** — 107 of a planned 130 patients; accrual stopped early for futility
- **n:** 107
- **Toxicities:** **neutropenia** (grade ≥ 3): 23% (n=52); **thrombocytopenia** (grade ≥ 3): 37% (n=52); **anemia** (grade ≥ 3): 13% (n=52); **hypertension** (grade 3): 8% (n=52); **thromboembolic event** (grade ≥ 3): 10% (n=52)
- **Efficacy:** **HR_PFS** 1.12; **PFS** 6.2 months; **PFS** 4.2 months; **HR_OS** 1.07; **ORR** 31.5%

Peposertib (DNA-PK inhibitor) + low-dose pegylated liposomal doxorubicin (NCT05711615), anthracycline-ceiling-gated

****Pautier et al. *Lancet Oncol* 2022****

- **Disease context:** **metastatic or unresectable leiomyosarcoma (uterine and soft-tissue), chemotherapy-naive (1L)** — ECOG 0-1; 67 of 150 patients had uterine LMS; randomisation stratified by uterine vs soft-tissue origin; surgery for residual disease allowed after 6 cycles
- **n:** 150

- **Toxicities:**neutropenia (grade ≥ 3): 80% (n=74); **thrombocytopenia** (grade ≥ 3): 47% (n=74); **anemia** (grade ≥ 3): 31% (n=74); **febrile neutropenia** (grade ≥ 3): 28% (n=74); **treatment-related death** (grade 5): 0.7% (n=150)
- **Efficacy:**PFS 12.2 months (95% CI 10.1-15.6); **PFS** 6.2 months (95% CI 4.1-7.1); **HR_PFS** 0.41 (95% CI 0.29-0.58)

All-trans retinoic acid + cemiplimab in leiomyosarcoma after standard chemotherapy (NCT06528769)

Ben-Ami et al. *Cancer* 2017

- **Disease context:**advanced uterine leiomyosarcoma, previously treated (2L+) — 12 patients, single centre; stage 1 of a 2-stage design; PD-L1 expression 20% of archival samples
- **n:** 12
- **Toxicities:**amylase/lipase increase (grade 4): 8% (n=12); **fatigue** (grade 3): 8% (n=12); **abdominal pain** (grade 3): 8% (n=12)
- **Efficacy:**ORR 0%; **PFS** 1.8 months

Zanzalitinib (XL092), next-generation multikinase inhibitor with a leiomyosarcoma cohort (NCT06571734)

van der Graaf et al. *Lancet* 2012

- **Disease context:**metastatic non-adipocytic soft-tissue sarcoma after standard chemotherapy (2L+) — 369 patients, angiogenesis-inhibitor-naïve; leiomyosarcoma was the largest histology group
- **n:** 369
- **Toxicities:**fatigue (grade any): 65%; **diarrhoea** (grade any): 58%; **nausea** (grade any): 54%; **weight loss** (grade any): 48%; **hypertension** (grade any): 41%
- **Efficacy:**HR_PFS 0.31 (95% CI 0.24-0.4); **PFS** 4.6 months (95% CI 3.7-4.8); **HR_OS** 0.86 (95% CI 0.67-1.11)

Pegylated liposomal doxorubicin maintenance after first-line chemotherapy (MELODY, NCT06981637)

Pautier et al. *Lancet Oncol* 2022

- **Disease context:**metastatic or unresectable leiomyosarcoma (uterine and soft-tissue), chemotherapy-naïve (1L) — ECOG 0-1; 67 of 150 patients had uterine LMS; randomisation stratified by uterine vs soft-tissue origin; surgery for residual disease allowed after 6 cycles
- **n:** 150
- **Toxicities:**neutropenia (grade ≥ 3): 80% (n=74); **thrombocytopenia** (grade ≥ 3): 47% (n=74); **anemia** (grade ≥ 3): 31% (n=74); **febrile neutropenia** (grade ≥ 3): 28% (n=74); **treatment-related death** (grade 5): 0.7% (n=150)
- **Efficacy:**PFS 12.2 months (95% CI 10.1-15.6); **PFS** 6.2 months (95% CI 4.1-7.1); **HR_PFS** 0.41 (95% CI 0.29-0.58)

Anlotinib (catequentinib, AL3818) in refractory soft-tissue sarcoma (APROMISS, NCT03016819)

Chi et al. *Clin Cancer Res* 2018

- **Disease context:**refractory metastatic soft-tissue sarcoma after anthracycline, angiogenesis-inhibitor-naïve (2L+) — 166 patients in China; LMS subset n=26, not uterine-specific
- **n:** 166
- **Toxicities:**hypertension (grade ≥ 3): 4.8% (n=166); **triglyceride elevation** (grade ≥ 3): 3.6% (n=166); **pneumothorax** (grade ≥ 3): 2.4% (n=166)

- **Efficacy:**ORR 13% (95% CI 7.6-18); **PFS** 11 months; **OS** 15 months; **PFS_rate** 75%

****Wang et al. *Clin Cancer Res* 2025****

- **Disease context:** —
- **n:** —
- **Toxicities:** —
- **Efficacy:** —

MASCT-I (autologous dendritic cell and cytokine-induced killer cell therapy) + doxorubicin + ifosfamide, first line (NCT06277154)

****Judson et al. *Lancet Oncol* 2014****

- **Disease context:**advanced or metastatic high-grade soft-tissue sarcoma, first line (1L) — age 18-60, WHO PS 0-1; all STS histologies, not LMS- or uterine-specific
- **n:** 455
- **Toxicities:**febrile neutropenia (grade ≥ 3): 46% (n=224); leucopenia (grade ≥ 3): 43% (n=224); neutropenia (grade ≥ 3): 42% (n=224); anemia (grade ≥ 3): 35% (n=224); thrombocytopenia (grade ≥ 3): 33% (n=224)
- **Efficacy:**OS 14.3 months (95% CI 12.5-16.5); **OS** 12.8 months (95% CI 10.5-14.3); **HR_OS** 0.83 (95% CI 0.67-1.03); **HR_PFS** 0.74 (95% CI 0.6-0.9); **ORR** 26%

Tislelizumab + liposomal doxorubicin and ifosfamide, first line (NCT06849986)

****Pollack et al. *JAMA Oncol* 2020****

- **Disease context:**advanced anthracycline-naïve sarcoma (LMS the most common subtype, 11 of 37) (1L) — 37 patients, single academic sarcoma centre; osteosarcoma, Ewing and alveolar/embryonal rhabdomyosarcoma excluded
- **n:** 37
- **Toxicities:** No dose-limiting toxicities or unexpected safety signals; tolerability of the chemo-IO combination was the main positive finding.
- **Efficacy:**ORR 19%; **PFS** 8.1 months (95% CI 7.6-10.8); **OS** 27.6 months

Nivolumab monotherapy in uterine leiomyosarcoma

****Ben-Ami et al. *Cancer* 2017****

- **Disease context:**advanced uterine leiomyosarcoma, previously treated (2L+) — 12 patients, single centre; stage 1 of a 2-stage design; PD-L1 expression 20% of archival samples
- **n:** 12
- **Toxicities:**amylase/lipase increase (grade 4): 8% (n=12); fatigue (grade 3): 8% (n=12); abdominal pain (grade 3): 8% (n=12)
- **Efficacy:**ORR 0%; **PFS** 1.8 months

****Tawbi et al. *Lancet Oncol* 2017****

- **Disease context:**advanced soft-tissue and bone sarcoma, up to 3 prior lines (2L+) — STS cohort n=40 evaluable, 10 per histology including 10 LMS
- **n:** 80
- **Toxicities:**anemia (grade ≥ 3): 7% (n=42); lymphopenia (grade ≥ 3): 7% (n=42); pneumonitis

(immune-related) (grade ≥ 3): 2% (n=84)

- **Efficacy:** ORR 18%; ORR 0%; ORR 40%

Nivolumab +/- ipilimumab, dual checkpoint blockade in sarcoma (Alliance A091401)

D'Angelo et al. *Lancet Oncol* 2018

- **Disease context:** metastatic sarcoma after ≥ 1 prior systemic line (2L+) — 85 treated across two non-comparative arms; LMS among enrolled histologies; responses in the combination arm included LMS and uterine sarcoma
- **n:** 85
- **Toxicities:** anemia (grade ≥ 3): 19% (n=42); hypotension (grade ≥ 3): 10% (n=42); **treatment-related serious AE** (grade any): 26% (n=42)
- **Efficacy:** ORR 16% (95% CI 7-30); ORR 5% (95% CI 1-16)

Ben-Ami et al. *Cancer* 2017

- **Disease context:** advanced uterine leiomyosarcoma, previously treated (2L+) — 12 patients, single centre; stage 1 of a 2-stage design; PD-L1 expression 20% of archival samples
- **n:** 12
- **Toxicities:** amylase/lipase increase (grade 4): 8% (n=12); fatigue (grade 3): 8% (n=12); abdominal pain (grade 3): 8% (n=12)
- **Efficacy:** ORR 0%; PFS 1.8 months

Doxorubicin + pembrolizumab, first-line chemo-immunotherapy (phase 1/2)

Pollack et al. *JAMA Oncol* 2020

- **Disease context:** advanced anthracycline-naïve sarcoma (LMS the most common subtype, 11 of 37) (1L) — 37 patients, single academic sarcoma centre; osteosarcoma, Ewing and alveolar/embryonal rhabdomyosarcoma excluded
- **n:** 37
- **Toxicities:** No dose-limiting toxicities or unexpected safety signals; tolerability of the chemo-IO combination was the main positive finding.
- **Efficacy:** ORR 19%; PFS 8.1 months (95% CI 7.6-10.8); OS 27.6 months

Pautier et al. *Lancet Oncol* 2022

- **Disease context:** metastatic or unresectable leiomyosarcoma (uterine and soft-tissue), chemotherapy-naïve (1L) — ECOG 0-1; 67 of 150 patients had uterine LMS; randomisation stratified by uterine vs soft-tissue origin; surgery for residual disease allowed after 6 cycles
- **n:** 150
- **Toxicities:** neutropenia (grade ≥ 3): 80% (n=74); thrombocytopenia (grade ≥ 3): 47% (n=74); anemia (grade ≥ 3): 31% (n=74); febrile neutropenia (grade ≥ 3): 28% (n=74); **treatment-related death** (grade 5): 0.7% (n=150)
- **Efficacy:** PFS 12.2 months (95% CI 10.1-15.6); PFS 6.2 months (95% CI 4.1-7.1); HR_PFS 0.41 (95% CI 0.29-0.58)

Checkpoint-blockade target biology in uterine leiomyosarcoma (immune classes, tertiary lymphoid structures, pS6-high sensitisation)

****Tawbi et al. *Lancet Oncol* 2017****

- **Disease context:** advanced soft-tissue and bone sarcoma, up to 3 prior lines (2L+) — STS cohort n=40 evaluable, 10 per histology including 10 LMS
- **n:** 80
- **Toxicities:** anemia (grade ≥ 3): 7% (n=42); lymphopenia (grade ≥ 3): 7% (n=42); pneumonitis (immune-related) (grade ≥ 3): 2% (n=84)
- **Efficacy:** ORR 18%; ORR 0%; ORR 40%

****Ben-Ami et al. *Cancer* 2017****

- **Disease context:** advanced uterine leiomyosarcoma, previously treated (2L+) — 12 patients, single centre; stage 1 of a 2-stage design; PD-L1 expression 20% of archival samples
- **n:** 12
- **Toxicities:** amylase/lipase increase (grade 4): 8% (n=12); fatigue (grade 3): 8% (n=12); abdominal pain (grade 3): 8% (n=12)
- **Efficacy:** ORR 0%; PFS 1.8 months

Ini1 Ihc interventions

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

Tazemetostat (EZH2 inhibition for SMARCB1/INI1-deficient disease) — Thin evidence

Access: Off-label

Likelihood of effect: Low even if INI1 is lost: 9% ORR in the INI1-negative non-epithelioid basket cohort, no uLMS patient reported (pmid:41882006).

Toxicity burden: Low (G3+ treatment-related AEs 10%; anaemia 6%, weight loss 3%; two serious events — one seizure, one haemoptysis)

Counter-productive MoA: **Moderate** (EZH2 dependency in SWI/SNF-mutant cancers is largely non-catalytic, so single-agent inhibition may not engage it.)

Overall: Gated on a stain the board expects to come back retained, and even a positive buys a 9% cross-histology response rate.

Key references: PMID 41882006 · PMID 33035459 · PMID 34271067

Valemetostat tosylate (EZH1/EZH2 dual inhibitor), SWI/SNF-defective basket cohort 1.A (NCT07303387) — Not enrollable

Access: Trial only

Likelihood of effect: Not enrollable: France-only, and the protocol requires biallelic homozygous SMARCB1

deletion plus central IHC confirmation that a heterozygous plasma cell cannot supply.

Toxicity burden: Moderate (EZH1/2 dual inhibition: thrombocytopenia, anaemia, dysgeusia in the lymphoma experience; no solid-tumour profile published)

Counter-productive MoA: **Moderate** (Same non-catalytic EZH2-dependency problem as tazemetostat, on a target the tissue may not carry.)

Overall: The live SWI/SNF basket that looks like a door and is not: biallelic homozygous deletion required, France only.

Key references: NCT07303387 · PMID 34271067

Tiragolumab + atezolizumab in SMARCB1- or SMARCA4-deficient tumours (NCT05286801) — _Not enrollable_

Access: Off-label

Likelihood of effect: Not enrollable: closed to accrual, and the TIGIT partner is investigational with no off-protocol route.

Toxicity burden: Moderate (checkpoint-class immune-related toxicity; combination G3+ profile not published for this study)

Counter-productive MoA: **Moderate** (Checkpoint blockade has failed twice in this histology; TIGIT adds no evidence in sarcoma.)

Overall: The most reachable SMARCB1-directed study for a US patient until it closed to accrual — worth re-checking, not plannable.

Key references: NCT05286801 · PMID 28988646

Ubamatamab (MUC16 x CD3 bispecific), alone or with cemiplimab, in SMARCB1-deficient tumours (NCT06444880) — _Not enrollable_

Access: Trial only

Likelihood of effect: Not enrollable: histology-restricted to renal medullary carcinoma and epithelioid sarcoma, and closed to accrual.

Toxicity burden: Moderate (T-cell-engager CRS risk; no published grade profile)

Counter-productive MoA: **Low** (No mechanism objection; the target antigen is unlikely to be present in this histology.)

Overall: A SMARCB1-platform study whose two cohorts exclude her histology, closed to accrual — checked and shut.

Key references: NCT06444880

BRD9 degraders (class) — ncBAF synthetic lethality in SMARCB1-loss disease — _Unavailable_

Access: Unavailable

Likelihood of effect: Not available: both clinical programmes terminated with no successor; no published efficacy from either.

Toxicity burden: Moderate (no case-relevant published toxicity data from either terminated programme)

Counter-productive MoA: **Moderate** (Preclinical work suggests the bromodomain is the wrong handle in SMARCB1-mutant disease.)

Overall: The second SMARCB1 approach, and it no longer exists — both clinical degraders terminated without a successor.

Key references: NCT04965753 · NCT05355753

FHD-609 (BRD9 degrader), phase 1 (NCT04965753) — _Unavailable_

Access: Unavailable

Likelihood of effect: Not available: phase 1 terminated, no successor study registered.

Toxicity burden: Moderate (no case-relevant published toxicity data)

Counter-productive MoA: **Moderate** (Class-level bromodomain-handle problem in SMARCB1-mutant disease.)

Overall: Terminated phase 1 in the SMARCB1-loss space, consolidated under the BRD9 class row.

Key references: NCT04965753 · PMID 39660994

CFT8634 (BRD9 degrader), phase 1 (NCT05355753) — _Unavailable_

Access: Unavailable

Likelihood of effect: Not available: phase 1 terminated, no follow-on programme registered.

Toxicity burden: Moderate (no case-relevant published toxicity data)

Counter-productive MoA: **Moderate** (Class-level bromodomain-handle problem in SMARCB1-mutant disease.)

Overall: The second BRD9 degrader to stop in SMARCB1-null disease — the class closed twice, not once.

Key references: NCT05355753 · PMID 41285423

Evidence in detail

Tazemetostat (EZH2 inhibition for SMARCB1/INI1-deficient disease)

Gounder et al. *Nat Commun* 2026

- **Disease context:** solid tumours with SWI/SNF alterations: rhabdoid tumours, synovial sarcoma, INI1-negative tumours of any histology, renal medullary carcinoma, poorly differentiated chordoma (2L+) — 129 patients across five cohorts; Cohort 3 (INI1-negative any histology, n=32) is the cohort a

tissue-confirmed INI1-lost uLMS would have entered

- **n:** 129
- **Toxicities:** any treatment-related AE (grade ≥ 3): 10% (n=129)
- **Efficacy:** ORR 9%; ORR 9%; ORR 6%; ORR 0%; PFS_rate 15.2%; TRAE_rate 10%

****Gounder et al. *Lancet Oncol* 2020****

- **Disease context:** advanced epithelioid sarcoma with INI1/SMARCB1 loss (any) — 62 patients, ECOG 0-2, documented INI1 loss by IHC or biallelic SMARCB1 alteration; 1L and later lines
- **n:** 62
- **Toxicities:** anemia (grade ≥ 3): 6% (n=62); weight loss (grade ≥ 3): 3% (n=62); treatment-related serious AE (grade ≥ 3): 3% (n=62)
- **Efficacy:** ORR 15% (95% CI 7-26); DCR 26% (95% CI 16-39); PFS 5.5 months (95% CI 3.4-5.9); OS 19.0 months; TTR 3.9 months

****Kihara et al. *Hum Pathol* 2021****

- **Disease context:** —
- **n:** —
- **Toxicities:** —
- **Efficacy:** —

Valemestostat tosylate (EZH1/EZH2 dual inhibitor), SWI/SNF-defective basket cohort 1.A (NCT07303387)

****Kihara et al. *Hum Pathol* 2021****

- **Disease context:** —
- **n:** —
- **Toxicities:** —
- **Efficacy:** —

Tiragolumab + atezolizumab in SMARCB1- or SMARCA4-deficient tumours (NCT05286801)

****Tawbi et al. *Lancet Oncol* 2017****

- **Disease context:** advanced soft-tissue and bone sarcoma, up to 3 prior lines (2L+) — STS cohort n=40 evaluable, 10 per histology including 10 LMS
- **n:** 80
- **Toxicities:** anemia (grade ≥ 3): 7% (n=42); lymphopenia (grade ≥ 3): 7% (n=42); pneumonitis (immune-related) (grade ≥ 3): 2% (n=84)
- **Efficacy:** ORR 18%; ORR 0%; ORR 40%

FHD-609 (BRD9 degrader), phase 1 (NCT04965753)

****Livingston et al. *Clin Cancer Res* 2025****

- **Disease context:** —
- **n:** —
- **Toxicities:** —
- **Efficacy:** —

CFT8634 (BRD9 degrader), phase 1 (NCT05355753)

** Jackson et al. *J Med Chem* 2025**

- **Disease context:** —
- **n:** —
- **Toxicities:** —
- **Efficacy:** —

Magea4 IHC interventions

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

CDR404 (MAGE-A4 x CD3 T-cell engager) in HLA-A*02:01, MAGE-A4-expressing solid tumours (NCT06402201) — _Thin evidence_

Access: Trial only

Likelihood of effect: Unknown: first-in-human phase 1 with a dose-limiting-toxicity endpoint and no efficacy readout; MAGE-A4 status unmeasured.

Toxicity burden: Moderate (T-cell-engager CRS risk; no published grade profile)

Counter-productive MoA:Low (Antigen-loss and HLA LOH escape possible; no data either way in this histology.)

Overall:An off-the-shelf MAGE-A4 engager with no manufacturing lead time and possibly no prior-line requirement, but zero efficacy data and an unstained antigen.

Key references: NCT06402201 · PMID 40152485

Afamitresgene autoleucel (MAGE-A4 TCR-T, TECELRA), SPEARHEAD-1 — _Not enrollable_

Access: Not yet accessible

Likelihood of effect: Not enrollable: ORR 37% is synovial sarcoma and MRCLS (pmid:38554725); the label stops at synovial sarcoma and her histology is outside it.

Toxicity burden: High (CRS 71% with one G3; G3+ lymphopenia 96%, neutropenia 85%; no treatment-related deaths)

Counter-productive MoA:Low (Antigen-loss escape possible; no mechanism objection, the barrier is histology and label.)

Overall:Her HLA type clears the allele gate and nothing else — the label and the trial both stop at synovial sarcoma.

Key references: PMID 38554725 · NCT04044768

Uzatresgene autoleucel (ADP-A2M4CD8), alone or with checkpoint blockade (SURPASS, NCT04044859) — _Not enrollable_

Access: Not yet accessible

Likelihood of effect: Not enrollable: closed to accrual and the histology list excludes uterine sarcoma.

Toxicity burden: High (TCR-T class: near-universal G3+ lymphodepletion cytopenias, CRS; no study-specific published profile)

Counter-productive MoA:Low (Antigen-loss escape possible; the operative barriers are histology, enrolment and an oximetry threshold.)

Overall:Closed, and its histology list excludes her — but its 90% room-air saturation criterion is the class gate her altitude makes decisive.

Key references: NCT04044859 · PMID 38554725

Cancer/testis antigen expression axis (MAGE-A4, PRAME, NY-ESO-1) — target biology behind the HLA-A*02:01 options — _Consolidated_

Access: Trial only

Likelihood of effect: Consolidated target biology, not a therapy. The prevalence base for smooth-muscle tumours is thin and internally inconsistent (pmid:40152485).

Toxicity burden: Low (none — this row is target biology; toxicity is carried on the product rows)

Counter-productive MoA:Moderate (HLA-A*02:01 allele loss would close the whole axis at once, whatever the antigen stains show.)

Overall:The stain-second half of a two-part gate: her allele is confirmed, all three antigens are untested, and the prevalence base is thin.

Key references: PMID 40152485 · PMID 38554725

Evidence in detail

Afamitresgene autoleucel (MAGE-A4 TCR-T, TECELRA), SPEARHEAD-1

****D'Angelo et al. *Lancet* 2024****

- **Disease context:**advanced synovial sarcoma or myxoid round cell liposarcoma, HLA-A*02, MAGE-A4-expressing, after anthracycline- or ifosfamide-containing therapy (2L+) — 52 patients (44 synovial sarcoma, 8 MRCLS); median 3 prior lines; MAGE-A4 expression and HLA-A*02 required at screening
- **n:** 52
- **Toxicities:**cytokine release syndrome (grade any): 71% (n=52); cytokine release syndrome (grade 3): 2% (n=52); lymphopenia (grade >=3): 96% (n=52); neutropenia (grade >=3): 85% (n=52); leukopenia (grade >=3): 81% (n=52)

- **Efficacy:**ORR 37% (95% CI 24-51); **ORR** 39% (95% CI 24-55); **ORR** 25% (95% CI 3-65)

Uzatresgene autoleucel (ADP-A2M4CD8), alone or with checkpoint blockade (SURPASS, NCT04044859)

D'Angelo et al. *Lancet* 2024

- **Disease context:**advanced synovial sarcoma or myxoid round cell liposarcoma, HLA-A*02, MAGE-A4-expressing, after anthracycline- or ifosfamide-containing therapy (2L+) — 52 patients (44 synovial sarcoma, 8 MRCLS); median 3 prior lines; MAGE-A4 expression and HLA-A*02 required at screening
- **n:** 52
- **Toxicities:**cytokine release syndrome (grade any): 71% (n=52); **cytokine release syndrome** (grade 3): 2% (n=52); **lymphopenia** (grade >=3): 96% (n=52); **neutropenia** (grade >=3): 85% (n=52); **leukopenia** (grade >=3): 81% (n=52)
- **Efficacy:**ORR 37% (95% CI 24-51); **ORR** 39% (95% CI 24-55); **ORR** 25% (95% CI 3-65)

Cancer/testis antigen expression axis (MAGE-A4, PRAME, NY-ESO-1) — target biology behind the HLA-A*02:01 options

D'Angelo et al. *Lancet* 2024

- **Disease context:**advanced synovial sarcoma or myxoid round cell liposarcoma, HLA-A*02, MAGE-A4-expressing, after anthracycline- or ifosfamide-containing therapy (2L+) — 52 patients (44 synovial sarcoma, 8 MRCLS); median 3 prior lines; MAGE-A4 expression and HLA-A*02 required at screening
- **n:** 52
- **Toxicities:**cytokine release syndrome (grade any): 71% (n=52); **cytokine release syndrome** (grade 3): 2% (n=52); **lymphopenia** (grade >=3): 96% (n=52); **neutropenia** (grade >=3): 85% (n=52); **leukopenia** (grade >=3): 81% (n=52)
- **Efficacy:**ORR 37% (95% CI 24-51); **ORR** 39% (95% CI 24-55); **ORR** 25% (95% CI 3-65)

B7H3 interventions

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

B7-H3-directed antibody-drug conjugates (MGC026, GSK5764227 / HS-20093) — Thin evidence

Access: Trial only

Likelihood of effect: Unknown: dose-escalation phase 1 programmes with no sarcoma efficacy readout; target prevalence is high (97% positive, pmid:39478506) but unmeasured here.

Toxicity burden: Moderate (ADC-class myelosuppression, interstitial lung disease risk; no sarcoma-specific grade profile published)

Counter-productive MoA:Low (Antigen-density threshold effects could blunt payload delivery; no mechanism objection raised by the board.)

Overall:The surface target most likely to be present, riding a stain that costs almost nothing — but every accessible programme is a phase 1 with no efficacy data.

Key references: NCT06242470 · NCT06551142 · PMID 39478506

CC-3 (B7-H3 x CD3 bispecific antibody) (NCT05999396) — _Not enrollable_

Access: Not yet accessible

Likelihood of effect: Not enrollable: single German site with colorectal-written eligibility; the sarcoma data behind it are preclinical.

Toxicity burden: Moderate (T-cell-engager CRS risk; no published grade profile)

Counter-productive MoA:Low (Antigen-density threshold effects possible; no mechanism objection raised.)

Overall:An interesting preclinical B7-H3 engager with one German site and colorectal-written eligibility — checked and closed.

Key references: NCT05999396 · PMID 39478506

Atrx Alt interventions

1. ATR inhibition on the ATRX / ALT hypothesis (berzosertib, tuvusertib class)

Access: Trial only

Likelihood of effect: Low: the hypothesis was tested in ATRX-mutant LMS with proven target engagement and patients progressed within 4 months (pmid:39453756).

Toxicity burden: Moderate (ATR-inhibitor class myelosuppression; monotherapy profile from the negative phase 2)

Counter-productive MoA:High (ALT-selective ATR sensitivity failed to replicate; selecting on ATRX rather than SLFN11 is selecting on the wrong marker.)

Overall:The one hypothesis in this case that has already been tested in her histology with the right biomarker and the right pharmacodynamics, and lost.

Key references: PMID 39453756 · NCT04807816

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

Tuvusertib (ATR inhibitor) + temozolomide, NCI dose escalation (NCT05691491) — _Thin evidence_

Access: Trial only

Likelihood of effect: Low: the monotherapy form of this hypothesis failed in ATRX-mutant LMS with target engagement proven on biopsy (pmid:39453756).

Toxicity burden: Moderate (ATR-inhibitor plus alkylator myelosuppression; no published combination grade profile)

Counter-productive MoA: **Moderate** (ALT-selective ATR sensitivity failed to replicate, and the sarcoma cohort testing it progressed within four months.)

Overall: The only reachable ATR seat, carrying a hypothesis this histology has already tested and answered in the negative.

Key references: NCT05691491 · PMID 39453756

Nyeso1 IHC interventions

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

Letetresgene autoleucel (lete-cel, NY-ESO-1 TCR-T), IGNYTE-ESO (NCT03967223) — _Not enrollable_

Access: Not yet accessible

Likelihood of effect: Not enrollable: synovial sarcoma and MRCLS only, closed to accrual; the citable efficacy is a 12-patient pilot (pmid:29891538).

Toxicity burden: High (G3+ AEs in all 12 pilot patients; lymphodepletion cytopenias, CRS in 5 of 12 with two G3)

Counter-productive MoA: **Low** (Antigen-loss escape possible; the barrier is histology restriction, and NY-ESO-1 is low across sarcoma subtypes.)

Overall: Histology-restricted and closed, on an antigen that stains low across sarcoma — order it only inside the shared panel.

Key references: NCT03967223 · PMID 29891538

Evidence in detail

Letetresgene autoleucel (lete-cel, NY-ESO-1 TCR-T), IGNYTE-ESO (NCT03967223)

D'Angelo et al. *Cancer Discov* 2018

- **Disease context:** metastatic synovial sarcoma, HLA-A*02, NY-ESO-1-expressing (2L+) — 12 patients in the

reported high-antigen, full-lymphodepletion cohort

- **n:** 12
- **Toxicities:**cytokine release syndrome (grade any): 42% (n=12); lymphopenia (grade ≥3): 100% (n=12); leukopenia (grade ≥3): 92% (n=12); neutropenia (grade ≥3): 83% (n=12); anemia (grade ≥3): 83% (n=12); febrile neutropenia (grade ≥3): 17% (n=12)
- **Efficacy:**ORR 50%

Hla A0201 interventions

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

Tebentafusp (gp100 x CD3 ImmTAC), HLA-A*02:01-restricted — Not enrollable

Access: Off-label

Likelihood of effect: Not enrollable: gp100 is a melanocytic antigen absent from leiomyosarcoma; the OS result is uveal-melanoma-specific (pmid:34551229).

Toxicity burden: Moderate (rash 83%, pyrexia 76%, pruritus 69% any-grade, front-loaded; discontinuation for AE 2%, no treatment-related deaths)

Counter-productive MoA:Low (Target antigen absent in this histology; the row is class precedent, not a therapeutic proposal.)

Overall:The approved ImmTAC her allele fits and her tumour's biology does not — class precedent, plus the strictest oximetry threshold in the file.

Key references: PMID 34551229 · NCT06942442

Evidence in detail

Tebentafusp (gp100 x CD3 ImmTAC), HLA-A*02:01-restricted

Nathan et al. *N Engl J Med* 2021

- **Disease context:**previously untreated metastatic uveal melanoma, HLA-A*02:01-positive (1L) — 378 HLA-A*02:01 patients randomised 2:1 vs investigator's choice (pembrolizumab, ipilimumab, or dacarbazine)
- **n:** 378
- **Toxicities:**rash (grade any): 83% (n=252); pyrexia (grade any): 76% (n=252); pruritus (grade any): 69% (n=252); treatment discontinuation for AE (grade any): 2% (n=252)
- **Efficacy:**HR_OS 0.51 (95% CI 0.37-0.71); HR_PFS 0.73 (95% CI 0.58-0.94)